DSA 3050A End Semester Exam

PIZZA SALES DATASET ANALYSIS PROJECT

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DATASET OVERVIEW

Dataset Source

Source: Pizza Sales Dataset - Kaggle

Description of the dataset

This is a synthetic dataset I obtained from Kaggle that describes pizza sales for a pizza place somewhere in the US. There are 32 different kinds of pizzas that fall into 4 different categories:

- classic
- chicken
- supreme
- veggie

Variables

The original dataset is a table with 49574 rows and 7 variables:

- **id:** The ID for the order, which consists of one or more pizzas at a given date and time.
- **date:** A character representation of the order date, expressed in the ISO 8601 date format (YYYY-MM-DD).

- **time:** A character representation of the order time, expressed as a 24-hour time the ISO 8601 extended time format (hh:mm:ss).
- **name:** The short name for the pizza.
- **size:** The size of the pizza, which can either be S, M, L, XL or XXL.
- **type:** The category or type of pizza, which can either be classic, chicken, supreme, or veggie.
- **price:** The price of the pizza and the amount that it sold for (in USD).

Details: Each pizza in the dataset is identified by a short name. The following listings provide the full names of each pizza and their main ingredients:

a. **Classic Pizzas:**
    - classic_dlx: The Classic Deluxe Pizza.
    - big_meat: The Big Meat Pizza.
    - pepperoni: The Pepperoni Pizza.
    - hawaiian: The Hawaiian Pizza.
    - pep_msh_pep: The Pepperoni, Mushroom, and Peppers Pizza.
    - ital_cpcllo: The Italian Capocollo Pizza.
    - napolitana: The Napolitana Pizza.
    - the_greek: The Greek Pizza.

b. **Chicken Pizzas:**
    - thai_ckn: The Thai Chicken Pizza.
    - bbq_ckn: The Barbecue Chicken Pizza.
    - southw_ckn: The Southwest Chicken Pizza.
    - cali_ckn: The California Chicken Pizza.
    - ckn_pesto: The Chicken Pesto Pizza.
    - ckn_alfredo: The Chicken Alfredo Pizza.

c. **Supreme Pizzas:**
    - brie_carre: The Brie Carre Pizza.
    - calabrese: The Calabrese Pizza.
    - soppressata: The Soppressata Pizza.
    - sicilian: The Sicilian Pizza.
    - ital_supr: The Italian Supreme Pizza.
    - peppr_salami: The Pepper Salami Pizza.
    - prsc_argla: The Prosciutto and Arugula Pizza.
    - spinach_supr: The Spinach Supreme Pizza.
    - spicy_ital: The Spicy Italian Pizza.

d. Vegetable Pizzas:

- mexicana: The Mexicana Pizza.
- four_cheese: The Four Cheese Pizza.
- five_cheese: The Five Cheese Pizza.
- spin_pesto: The Spinach Pesto Pizza.
- veggie_veg: The Vegetables Pizza.
- green_garden: The Green Garden Pizza.
- mediterraneo: The Mediterranean Pizza.
- spinach_fet: The Spinach and Feta Pizza.
- ital_veggie: The Italian Vegetables Pizza.

Screenshots of the original dataset:

#	# X	id	date	time	name	size	type	# price
1	1	2015-000001	2015-01-01	11:38:36	hawaiian	M	classic	13.25
2	2	2015-000002	2015-01-01	11:57:40	classic_dlx	M	classic	16
3	3	2015-000002	2015-01-01	11:57:40	mexicana	M	veggie	16
4	4	2015-000002	2015-01-01	11:57:40	thai_ckn	L	chicken	20.75

Part A: Data Acquisition and Understanding

1. **Domain and Business Problem**

   Domain: Retail / Food & Beverage Analytics.

   Business Problem: Plato's Pizza, a high-volume artisanal pizzeria, is facing challenges in optimizing its operational efficiency and maximizing profitability. Despite steady sales, the management lacks a data-driven understanding of their peak periods, underperforming menu items, and customer purchasing patterns.

The ***primary objective of this project*** is to develop a comprehensive Power BI solution that transforms raw transactional data into actionable insights. This will enable stakeholders to:

- Identify sales trends across different time dimensions (monthly, daily, hourly).
- Optimize inventory management by identifying the most and least popular pizza categories and sizes.
- Increase Average Order Value (AOV) through data-backed promotional strategies and menu engineering.

2. **Suitability for Advanced Analysis**

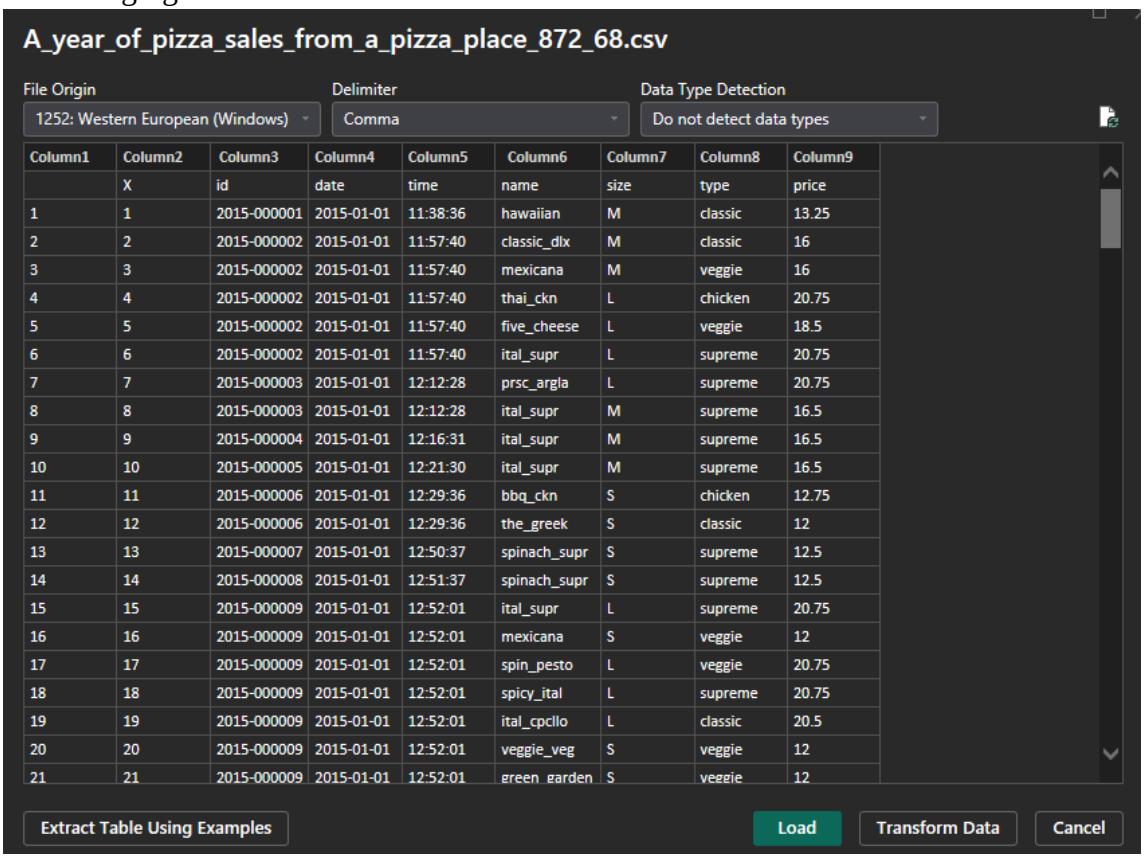
   This dataset meets the requirements because:

   - ***Relational Complexity:*** After adding tables, it will contain 4 interrelated tables, allowing for an appropriate Star Schema design.
   - ***Volume:*** With 48,620 rows, it exceeds the 9,000-row requirement threshold, ensuring statistical significance in the analysis.
   - ***Analytical Depth:*** The mix of categorical (Category, Size) and numerical (Price, Quantity) variables enables complex DAX calculations later on.
   - ***Time Intelligence:*** The dual date and time fields allow for granular *Peak Hour* analysis and traditional *Year-over-Year* reporting.

Part B: Data Cleaning and Transformation in Power Query

1. **Import and Initial Assessment**

   I imported the pizza sales dataset into Power BI using the Text/CSV connector.
   The raw data contained 49,574 transaction records with 9 columns, including two artifact index columns ("Unnamed: 0" and "X") that were generated during CSV export.
   Initial assessment showed no missing values, no duplicate records, and no data quality issues such as negative prices or invalid dates.
   The data spans the complete year 2015, from January 1 to December 31, with transaction times ranging from 09:52 to 23:05.

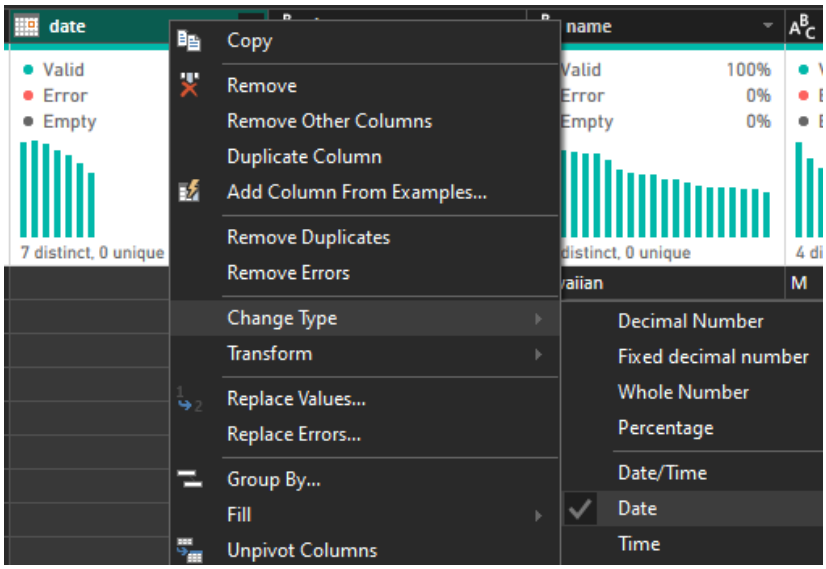


Column1	Column2	Column3	Column4	Column5	Column6	Column7	Column8	Column9
	X	id	date	time	name	size	type	price
1	1	2015-000001	2015-01-01	11:38:36	hawaiian	M	classic	13.25
2	2	2015-000002	2015-01-01	11:57:40	classic_dlx	M	classic	16
3	3	2015-000002	2015-01-01	11:57:40	mexicana	M	veggie	16
4	4	2015-000002	2015-01-01	11:57:40	thai_ckn	L	chicken	20.75
5	5	2015-000002	2015-01-01	11:57:40	five_cheese	L	veggie	18.5
6	6	2015-000002	2015-01-01	11:57:40	ital_supr	L	supreme	20.75
7	7	2015-000003	2015-01-01	12:12:28	prsc_argla	L	supreme	20.75
8	8	2015-000003	2015-01-01	12:12:28	ital_supr	M	supreme	16.5
9	9	2015-000004	2015-01-01	12:16:31	ital_supr	M	supreme	16.5
10	10	2015-000005	2015-01-01	12:21:30	ital_supr	M	supreme	16.5
11	11	2015-000006	2015-01-01	12:29:36	bbq_ckn	S	chicken	12.75
12	12	2015-000006	2015-01-01	12:29:36	the_greek	S	classic	12
13	13	2015-000007	2015-01-01	12:50:37	spinach_supr	S	supreme	12.5
14	14	2015-000008	2015-01-01	12:51:37	spinach_supr	S	supreme	12.5
15	15	2015-000009	2015-01-01	12:52:01	ital_supr	L	supreme	20.75
16	16	2015-000009	2015-01-01	12:52:01	mexicana	S	veggie	12
17	17	2015-000009	2015-01-01	12:52:01	spin_pesto	L	veggie	20.75
18	18	2015-000009	2015-01-01	12:52:01	spicy_ital	L	supreme	20.75
19	19	2015-000009	2015-01-01	12:52:01	ital_cpclio	L	classic	20.5
20	20	2015-000009	2015-01-01	12:52:01	veggie_veg	S	veggie	12
21	21	2015-000009	2015-01-01	12:52:01	green_garden	S	veggie	12

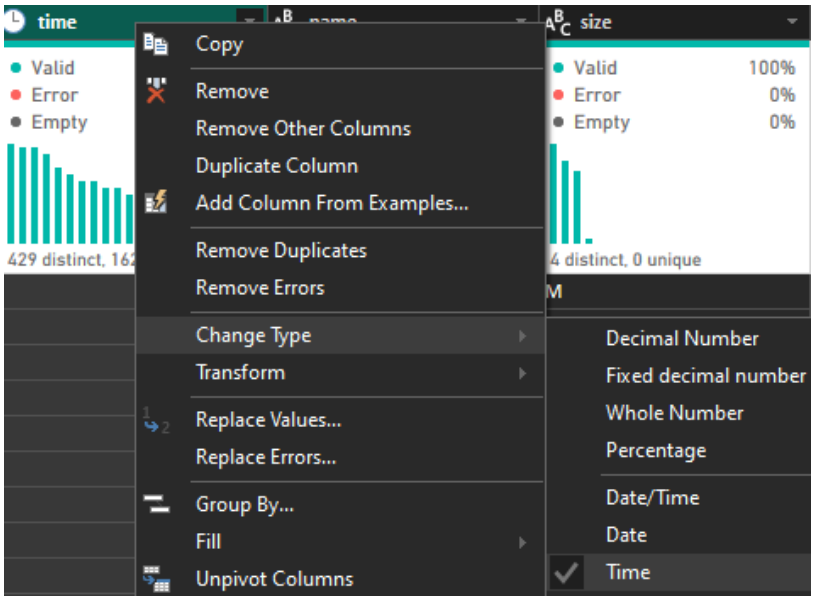
2. **Data Type Corrections**

   I applied data type corrections to ensure analytical validity:

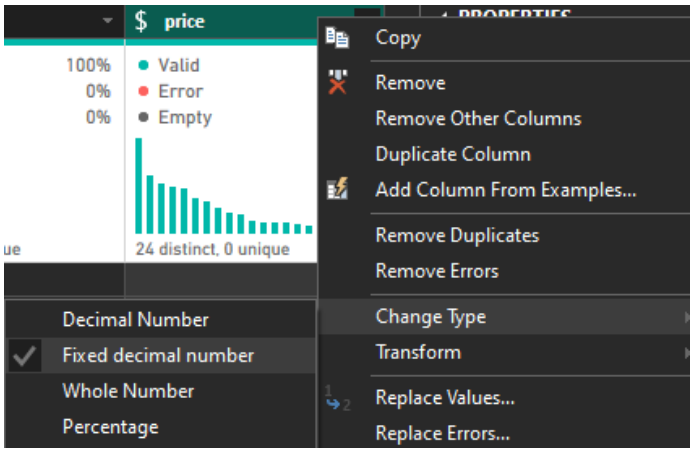
   a. ***Date Column:*** Changed from Text to Date type to enable time intelligence functions such as year-over-year comparisons and monthly aggregations.



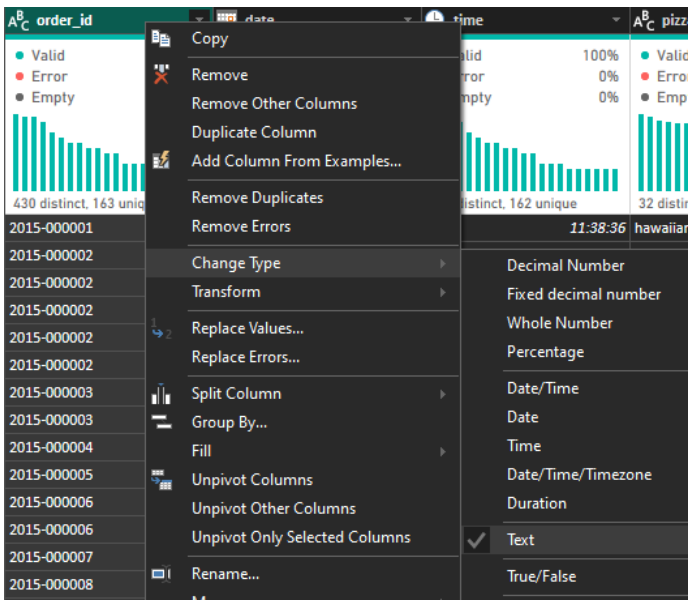
b. ***Time Column:*** Changed from Text to Time type to support hour-based analysis for identifying peak operational periods.



c. ***Price Column:*** Changed to Fixed Decimal Number (Currency) to maintain precision in financial calculations and ensure proper formatting in reports.



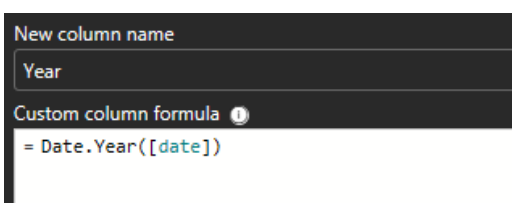
d. ***Order_id Column:*** Changed to Text type to prevent mathematical operations on identifier values and preserve leading zeros if present.



3. Derived Column Creation

I created derived columns to support dimensional analysis:

a. **Year**: I extracted it from date using Power Query's Date.Year function to enable annual trend analysis.



b. **Month**: I extracted it using Date.Month function for monthly performance tracking.

New column name

Month

Custom column formula

```
= Date.Month([date])
```

c. **Quarter**: I extracted it using Date.QuarterOfYear function for quarterly business reporting.

New column name

Quarter

Custom column formula

```
= Date.QuarterOfYear([date])
```

d. **Day of Week**: I extracted it using Date.DayOfWeekName function to identify weekday versus weekend patterns.

New column name

Day of Week

Custom column formula

```
= Date.DayOfWeek([date])
```

e. **Hour**: I extracted it using Time.Hour function from the time column to analyze peak sales periods.

New column name

Hour

Custom column formula

```
= Time.Hour([time])
```

f. **Revenue**: I created it as a calculated column in the fact table. Quantity is implicit as 1 in this dataset.

New column name

Revenue

Custom column formula

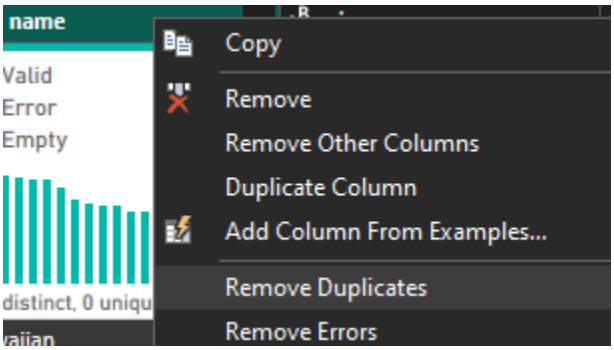
```
= [price] * 1
```

4. **Star Schema Construction**
   I transformed the flat transactional data into a Star Schema consisting of one fact table and three dimension tables:

Fact_Sales Table: Contains 49,574 transaction records with grain at the individual pizza sale level. Includes foreign keys to all dimensions (order_id, date, pizza_name, size) and measures (price, Revenue).

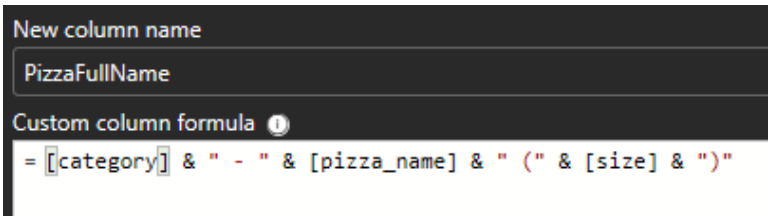
Dim_Pizza Table: Created by referencing the main query, selecting only pizza attributes (pizza_name, category, size).

I also removed duplicates from the pizza_name column. This resulted in 32 unique pizza combinations.

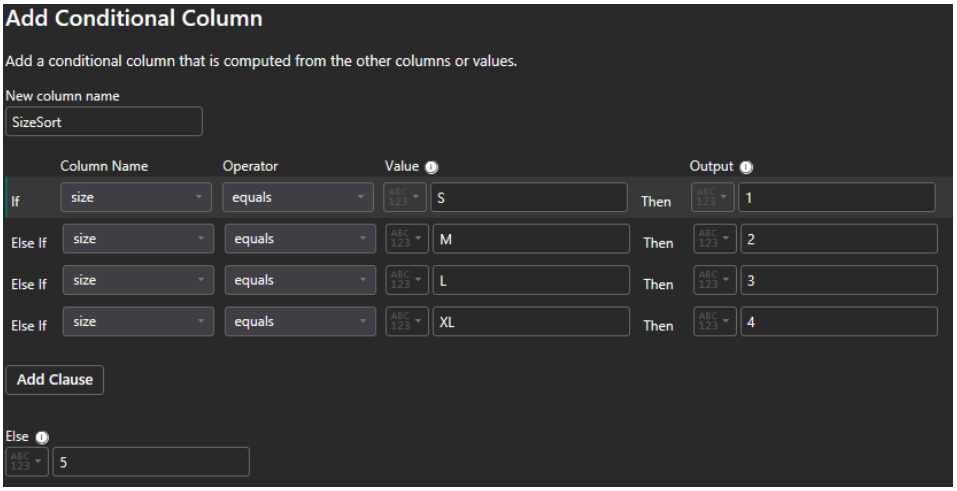


Two calculated columns were added:

- PizzaFullName: Concatenates category, pizza name, and size for display purposes.

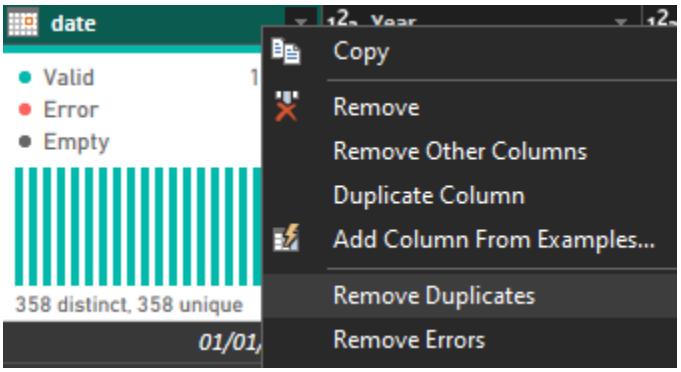


- SizeSort: Numeric sort key (1-5) to ensure proper size ordering (S, M, L, XL, XXL).



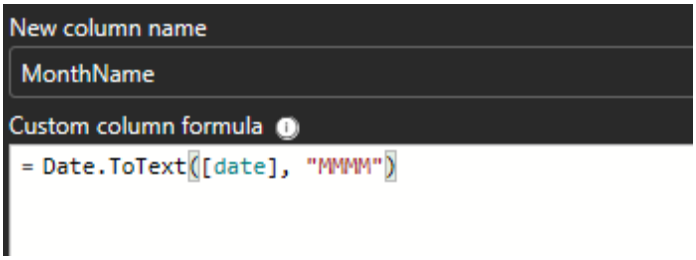
Dim_Date Table: I created it by referencing date-related columns.

I also removed duplicates to produce a calendar table with 358 unique dates.

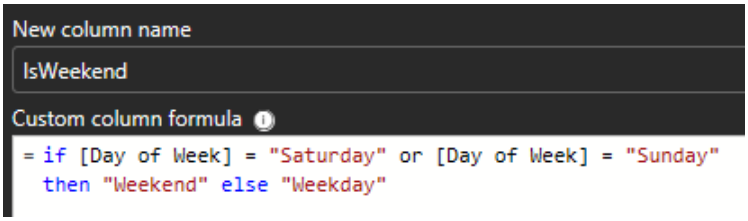


I then created additional calculated columns:

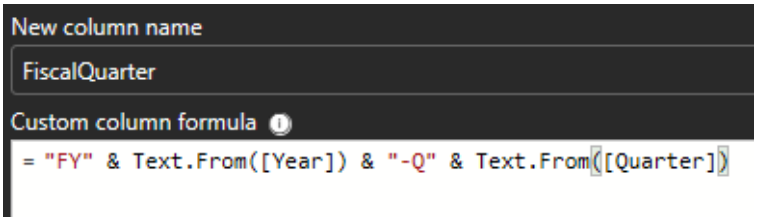
- MonthName: Full month name for report display.



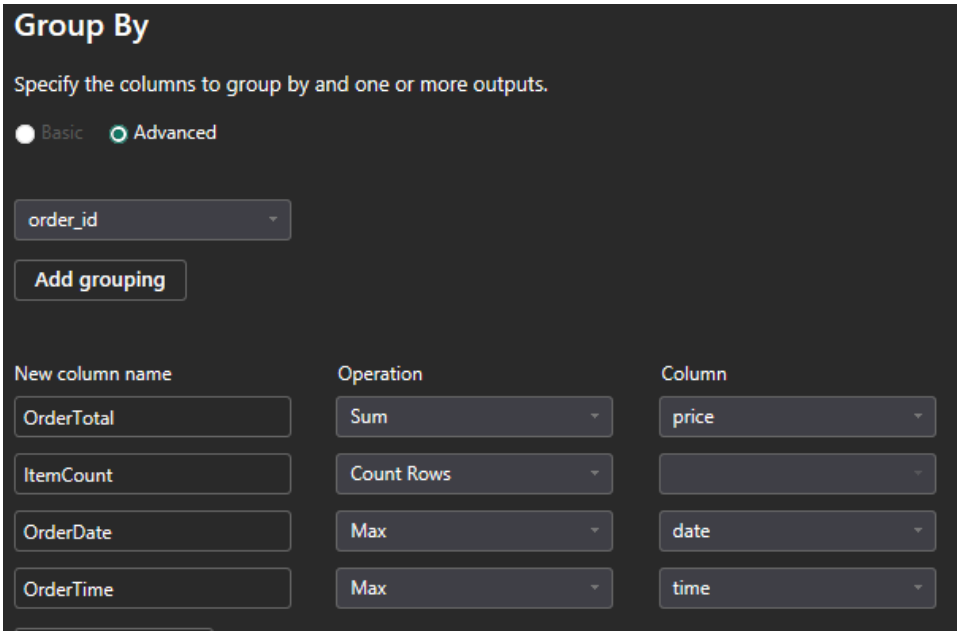
- IsWeekend: Binary classification of weekend versus weekday.



- FiscalQuarter: Formatted quarter identifier (FY2015-Q1, etc.).



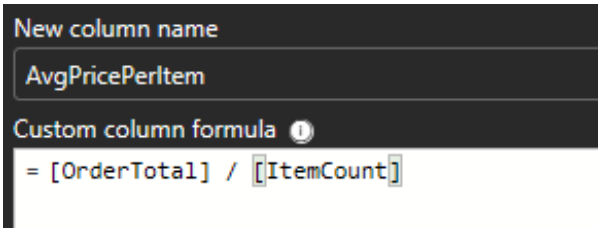
***Dim_Orders Table**:* I created it through aggregation of line items to the order level using the Group By function. Grouped by order_id with three aggregations:



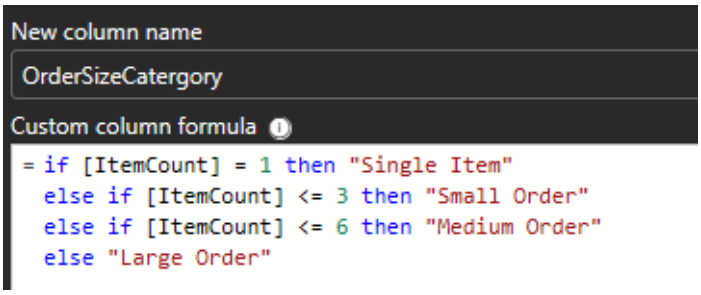
- OrderTotal: Sum of all item prices in the order.
- ItemCount: Count of pizzas in the order.
- OrderDate: Maximum date (all items in order share same date).
- OrderTime: Maximum time.

Two calculated columns were added:

- **AvgPricePerItem:** OrderTotal divided by ItemCount.

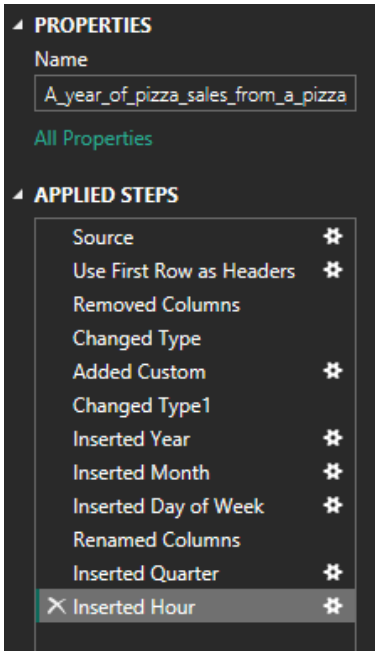


- **OrderSizeCategory*:*** Categorization as *Single Item*, *Small Order* (2-3 items), *Medium Order* (4-6 items), or *Large Order* (7+ items).

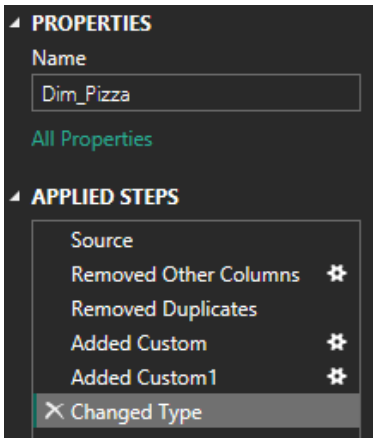


5. **Transformation Summary**
   The transformation steps documented in the Power Query Applied Steps pane:

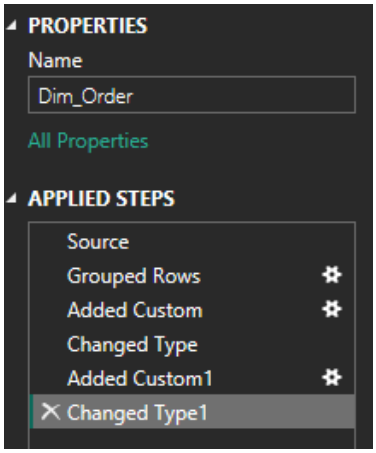
   a. ***Main Table:***



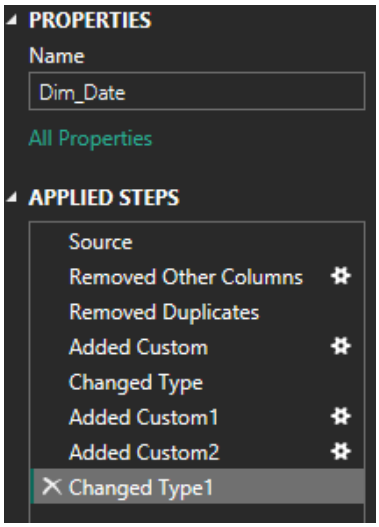
b. Dim_Pizza:



c. Dim_Orders:



d. Dim_Date:



Step	Transformation	Purpose
1	Removed "Unnamed" column	Eliminate artifact index
2	Removed "X" column	Eliminate duplicate index
3	Renamed columns	Standardize naming convention
4	Changed date to Date type	Enable time intelligence
5	Changed time to Time type	Enable hour analysis
6	Changed price to Fixed Decimal	Ensure financial precision
7	Extracted Year, Month, Quarter	Time dimension attributes
8	Extracted DayOfWeek and Hour	Operational analysis support
9	Created Dim_Pizza with deduplication	Product dimension
10	Created Dim_Date with calendar logic	Date dimension
11	Created Dim_Orders with Group By	Order-level aggregation
12	Added calculated columns	Business logic implementation

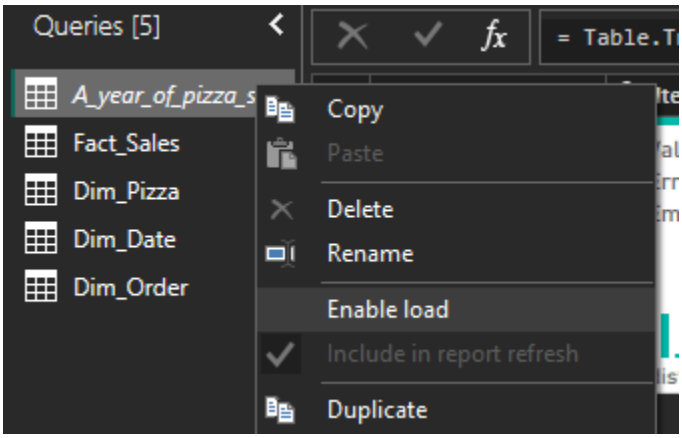
Rationale for Star Schema Design

I selected Star Schema for this project because:

1. ***Query Performance:*** The dimensional model reduces data redundancy and enables efficient filtering through relationships.

2. ***Analytical Flexibility****:* Separating dimensions (pizza attributes, time, orders) from facts (transactions) allows flexible slicing and dicing of data.
3. ***DAX Compatibility****:* Time intelligence functions in DAX require properly structured date tables with continuous date ranges.
4. ***Business Alignment****:* The schema mirrors business concepts such as pizzas as products, dates as time context, orders as customer transactions, and sales as measurable events.

The intermediate query (A_year_of_pizza_sales_from_a_pizza_place_872_68) containing all raw transformations was disabled from loading to the model to reduce memory consumption and prevent confusion, ensuring only the four production tables (Fact_Sales, Dim_Pizza, Dim_Date, Dim_Orders) are available for reporting.



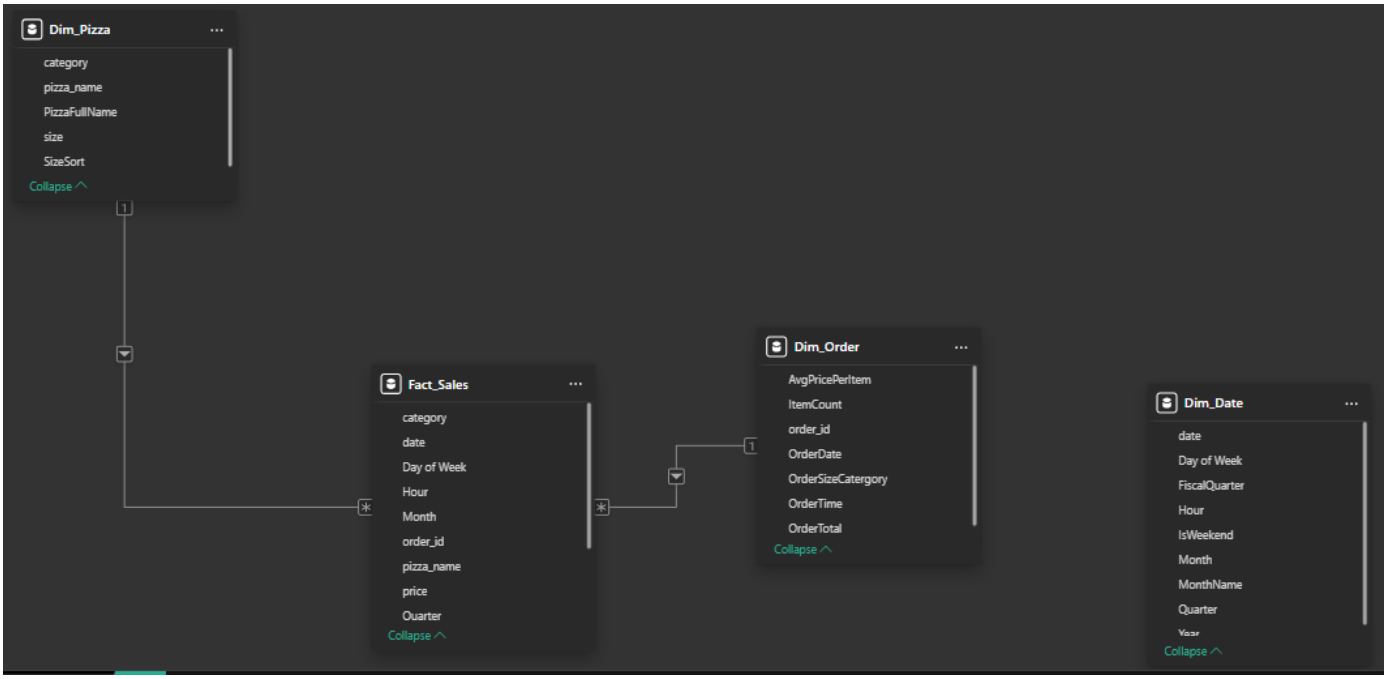
Part C: Data Modeling

1. **Model Architecture**
   I implemented a Star Schema consisting of one fact table surrounded by three dimension tables:

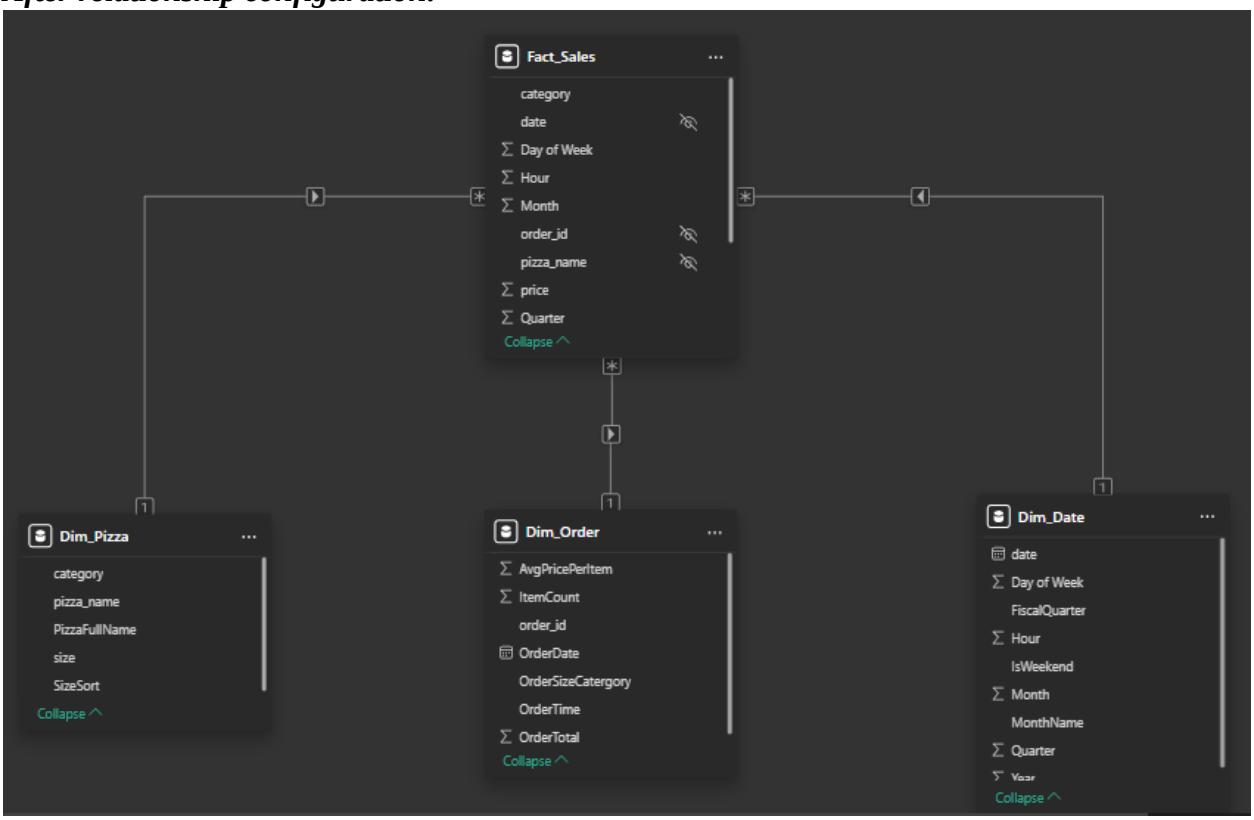
   Table Roles and Business Purpose

Table Name	Role	Primary Key	Business Purpose
Fact_Sales	Fact Table	order_details_id	Contains all transactional events (49,574 records). Contains measurable metrics (price, revenue) and foreign keys to dimensions.
Dim_Pizza	Dimension Table	pizza_name	Descriptive attributes of products (32 unique pizzas). Enables analysis by category, size, and full product name.
Dim_Date	Dimension Table	date	Time intelligence support (358 unique dates). Contains derived time attributes (Year, Month, Quarter, DayOfWeek, IsWeekend).
Dim_Orders	Dimension Table	order_id	Order-level aggregation (21,350 unique orders). Enables analysis of customer purchasing behavior (order size, average items).

2. **Relationship Configuration**
   Before Relationship configuration:

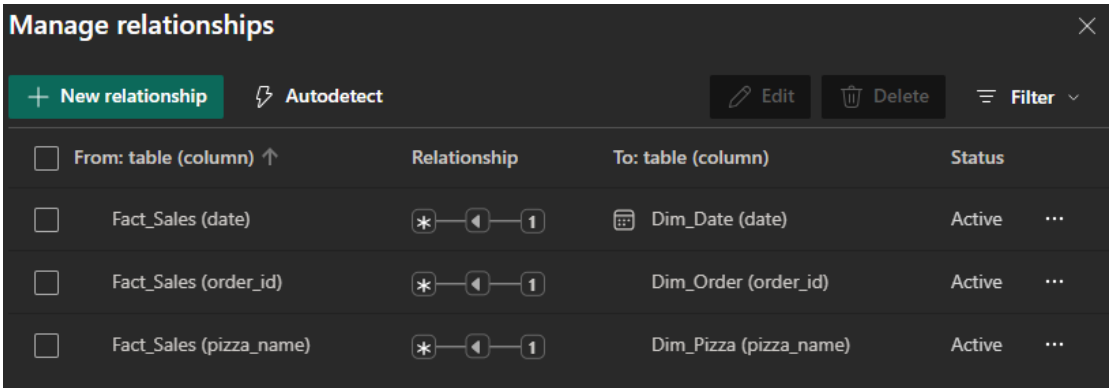


After relationship configuration:



Relationship Details:

From Table	To Table	Column	Cardinality	Cross-Filter Direction	Active
Fact_Sales	Dim_Pizza	pizza_name	Many-to-One (:1)	Single (Dim_Pizza → Fact_Sales)	Yes
Fact_Sales	Dim_Date	date	Many-to-One (:1)	Single (Dim_Date → Fact_Sales)	Yes
Fact_Sales	Dim_Orders	order_id	Many-to-One (:1)	Single (Dim_Orders → Fact_Sales)	Yes



Rationale for Relationship Choices:

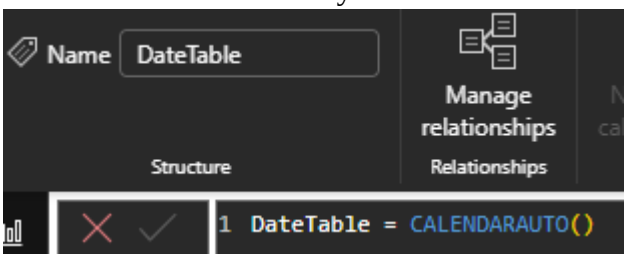
a. **Cardinality (Many-to-One):**
   - Multiple sales transactions can involve the same pizza, same date, or same order ID.
   - This is the standard pattern for dimensional modeling where facts reference dimensions.

b. **Cross-Filter Direction (Single):**
   - Prevents filter ambiguity and circular reference errors.
   - Dimensions filter facts, not vice versa, ensuring clean aggregation behavior.
   - Example: Selecting Classic category in Dim_Pizza filters Fact_Sales to show only classic pizza sales.

c. **Active Relationships:**
   - All three relationships are active, enabling simultaneous filtering across all dimensions.
   - No inactive relationships required as the model follows pure star schema design.

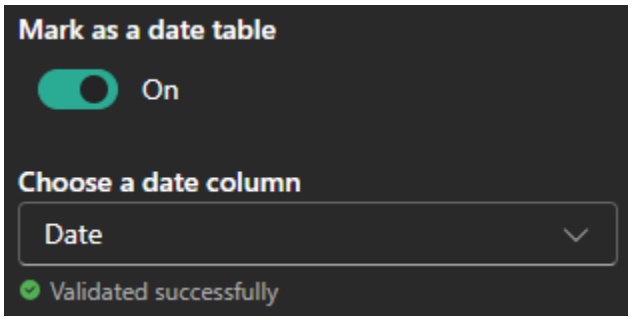
3. **Date Table Configuration**

I created a new calculated table called DateTable which generates every single day in my range of dates from the ‘date’ column. This table is necessary since I cannot mark Dim_Date as an official date table since the ‘date’ column has gaps because it only contains dates with activity.

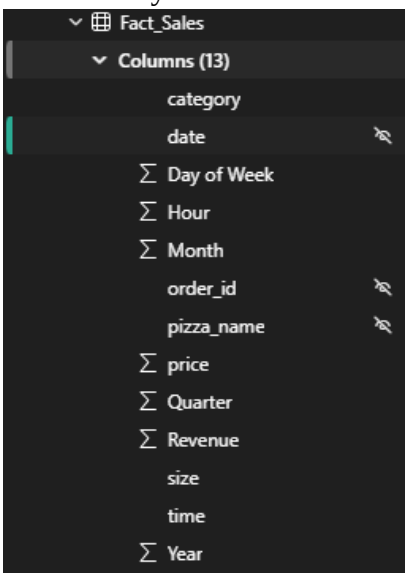


I marked DateTable as an official date table through the following process:

- Right-clicked DateTable → "Mark as date table".
- Selected the 'Date' column as the unique identifier.
- This enables Power BI's built-in time intelligence functions (SAMEPERIODLASTYEAR, DATESYTD, etc.).



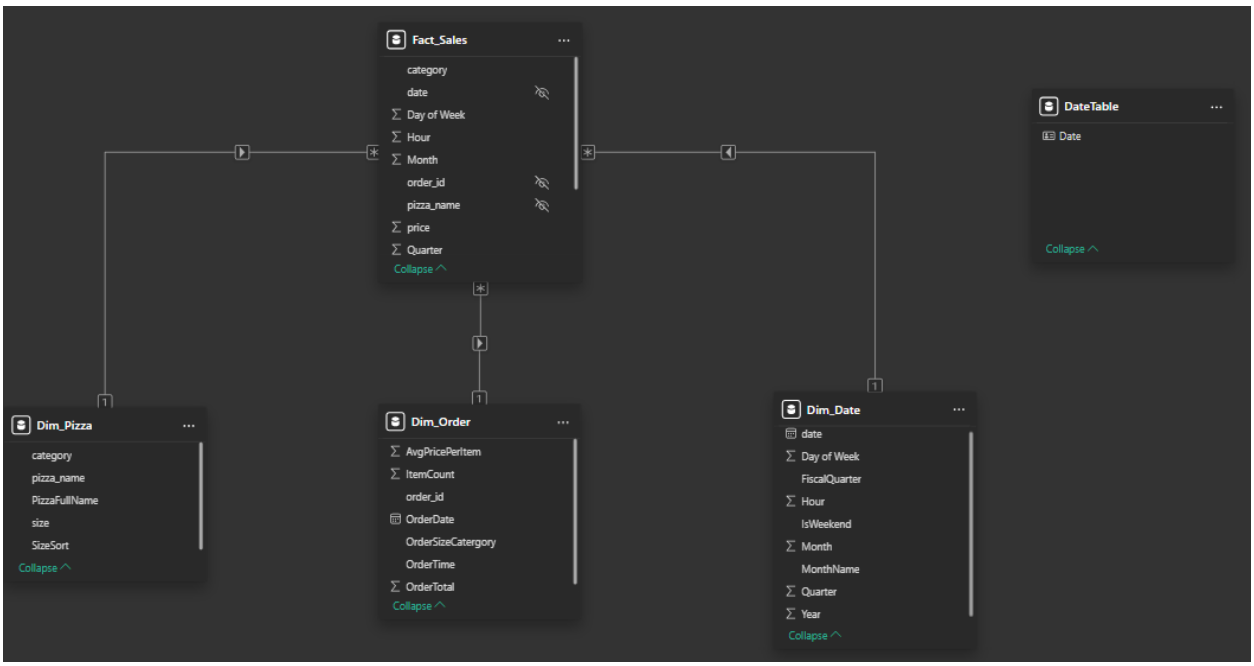
4. **Model Optimization**
   Hidden Columns: Foreign key columns (pizza_name, date, order_id) in Fact_Sales are hidden from report view to prevent end-user confusion while maintaining relationship functionality.



5. **Why This Model Supports Efficient Reporting**

Design Feature	Reporting Benefit
Star Schema	Simplifies DAX calculations; filters flow naturally from dimensions to facts
Separate Dim_Orders	Enables order-level metrics (AOV, order size distribution) independent of line-item grain
Dedicated Date Dimension	Supports time intelligence without complex DAX; continuous date range ensures no gaps
SizeSort Column	Guarantees proper size ordering (S→M→L→XL→XXL) in visualizations
IsWeekend Flag	Instant weekday vs. weekend analysis capability

Final Model View:

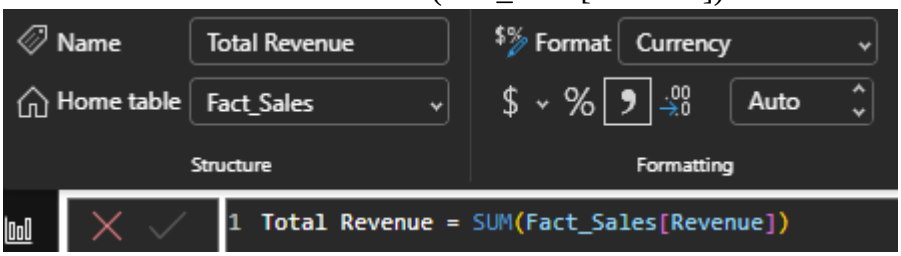


Part D: DAX Measures and Calculated Columns

DAX Measures Created

1. **Total Revenue**

   Formula: Total Revenue = SUM(Fact_Sales[Revenue])

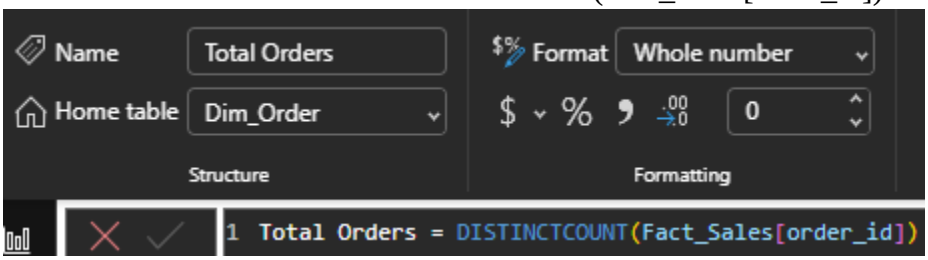
   

   Purpose: Calculates the sum of all pizza sales revenue across the entire dataset or any filtered subset. This is the foundational measure for all financial analysis.

   Decision Support: Enables instant visibility into total business performance and is used as baseline for all percentage and comparison calculations.

2. **Total Orders**

   Formula: Total Orders = DISTINCTCOUNT(Fact_Sales[order_id])

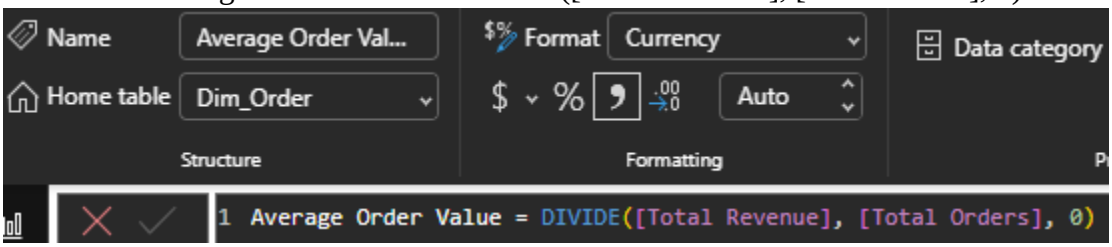
   

   Purpose: Counts unique order identifiers to determine how many separate customer transactions occurred.

   Decision Support: Critical for understanding customer traffic patterns It is combined with revenue to calculate average order value.

3. **Average Order Value (AOV)**

   Formula: Average Order Value = DIVIDE([Total Revenue], [Total Orders], 0)

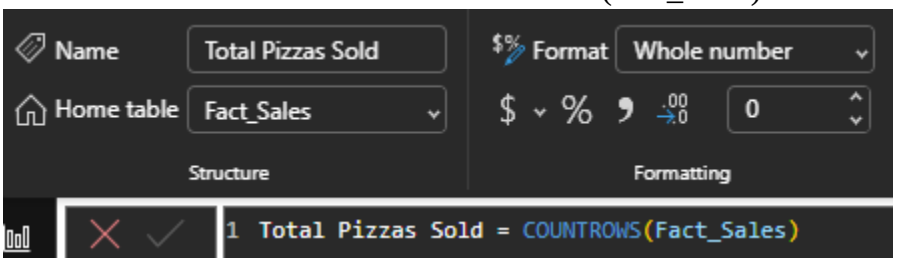
   

   Purpose: Computes the mean revenue per transaction, handling division-by-zero scenarios.

   Decision Support: Key metric for pricing strategy evaluation. Increasing AOV is often more cost-effective than acquiring new customers.

4. **Total Pizzas Sold**

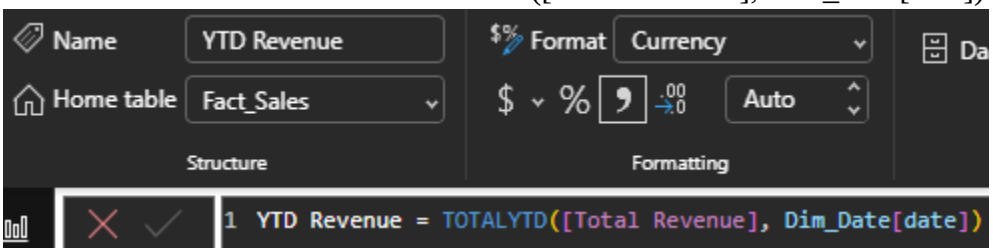
Formula: Total Pizzas Sold = COUNTROWS(Fact_Sales)



Purpose: Counts individual pizza items sold (line-item level), distinct from order count.
Decision Support: Operational metric for kitchen capacity planning and inventory forecasting; reveals upselling effectiveness.

5. **Year-to-Date Revenue**

Formula: YTD Revenue = TOTALYTD([Total Revenue], Dim_Date[date])

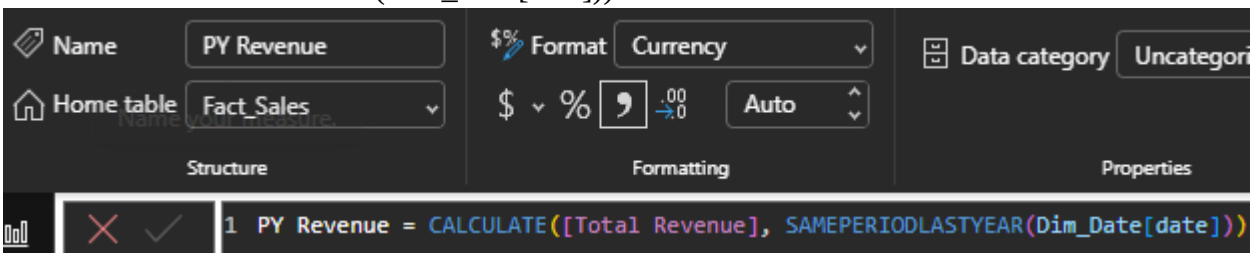


Purpose: Accumulates revenue from January 1st through the current filter context date.
Decision Support: Essential for tracking annual goal progress; enables comparison against targets at any point in the year.

6. **Previous Year Revenue**

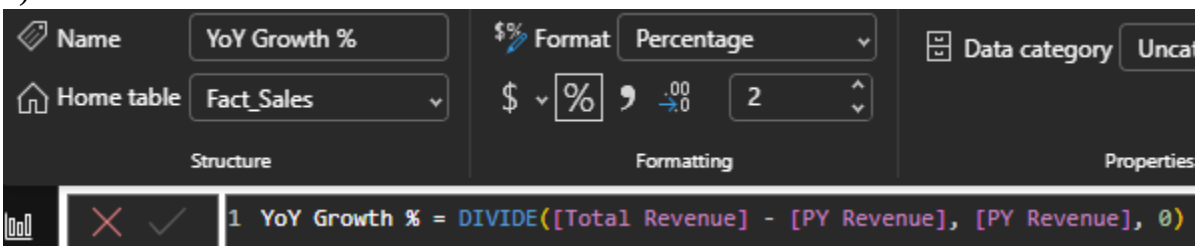
Formula:

PY Revenue = CALCULATE([Total Revenue], SAMEPERIODLASTYEAR(Dim_Date[date]))



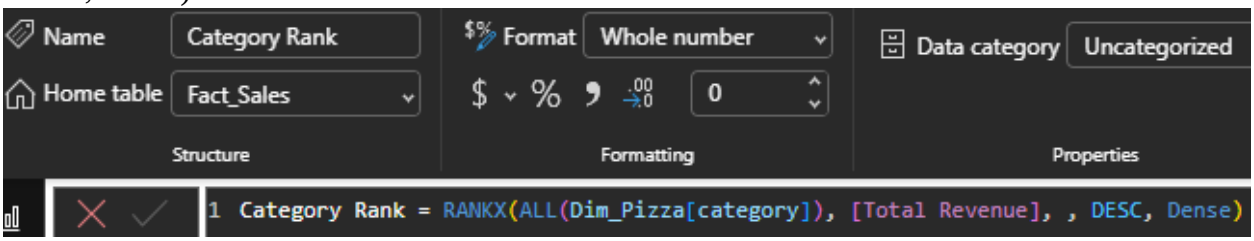
Purpose: Retrieves revenue from the same period in the prior year using time intelligence.
Decision Support: Baseline for growth analysis; contextualizes current performance against historical benchmarks.

7. **Year-over-Year Growth %**
   Formula: YoY Growth % = DIVIDE([Total Revenue] - [PY Revenue], [PY Revenue], 0)



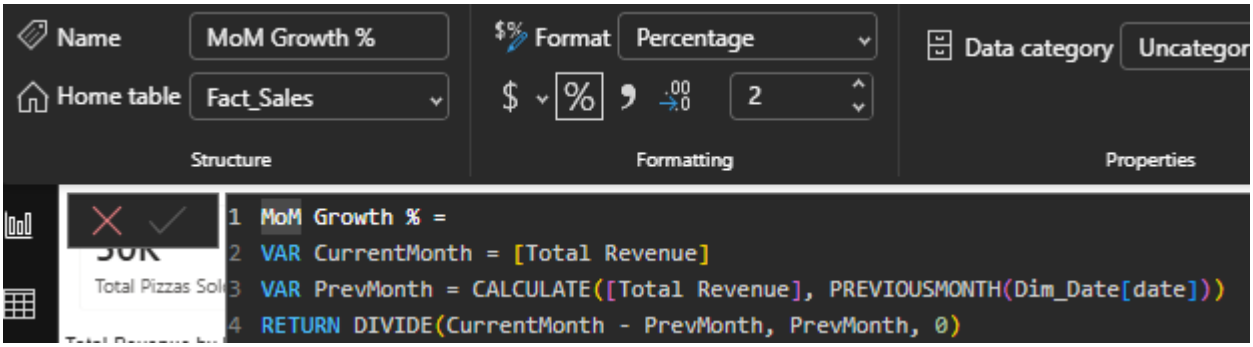
   Purpose: Calculates percentage change between current and prior year periods.
   Decision Support: Standardized performance indicator; positive values indicate growth, negative values signal decline requiring investigation.

8. **Category Rank**
   Formula: Category Rank = RANKX(ALL(Dim_Pizza[category]), [Total Revenue], , DESC, Dense)



   Purpose: Dynamically ranks pizza categories (Classic, Chicken, Supreme, Veggie) by revenue performance.
   Decision Support: Identifies top and bottom performing categories instantly; supports resource allocation decisions (promote winners, fix losers).

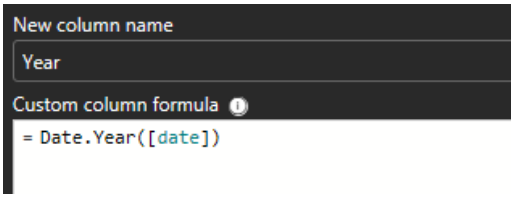
9. **Month-over-Month**
   Formula:
   MoM Growth % =
   VAR CurrentMonth = [Total Revenue]
   VAR PrevMonth = CALCULATE([Total Revenue], PREVIOUSMONTH(Dim_Date[date]))
   RETURN DIVIDE(CurrentMonth - PrevMonth, PrevMonth, 0)



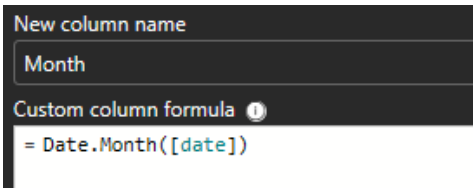
Calculated Columns Created

I had created the following calculated columns in Power Query as discussed in part B:

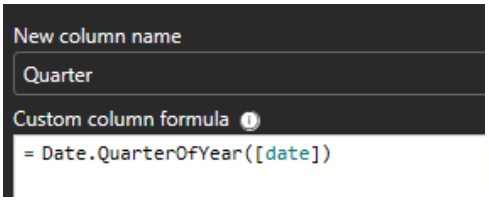
1. **Year**: I extracted it from date using Power Query's Date.Year function to enable annual trend analysis.



2. **Month**: I extracted it using Date.Month function for monthly performance tracking.



3. **Quarter**: I extracted it using Date.QuarterOfYear function for quarterly business reporting.



4. **Day of Week**: I extracted it using Date.DayOfWeekName function to identify weekday versus weekend patterns.

New column name

Day of Week

Custom column formula

```
= Date.DayOfWeek([date])
```

5. **Hour**: I extracted it using Time.Hour function from the time column to analyze peak sales periods.

New column name

Hour

Custom column formula

```
= Time.Hour([time])
```

6. **Revenue**: I created it as a calculated column in the fact table. Quantity is implicit as 1 in this dataset.

New column name

Revenue

Custom column formula

```
= [price] * 1
```

7. **MonthName:** Full month name for report display.

New column name

MonthName

Custom column formula

```
= Date.ToText([date], "MMMM")
```

8. **IsWeekend:** Binary classification of weekend versus weekday.

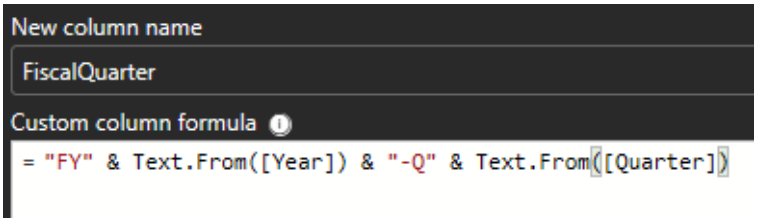
New column name

IsWeekend

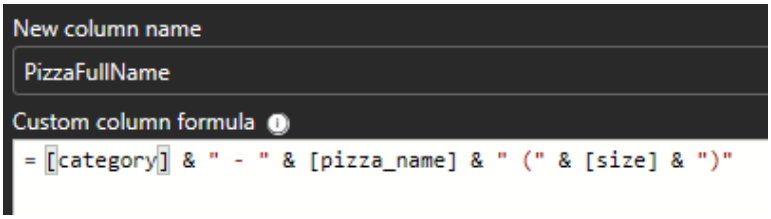
Custom column formula

```
= if [Day of Week] = "Saturday" or [Day of Week] = "Sunday"
  then "Weekend" else "Weekday"
```

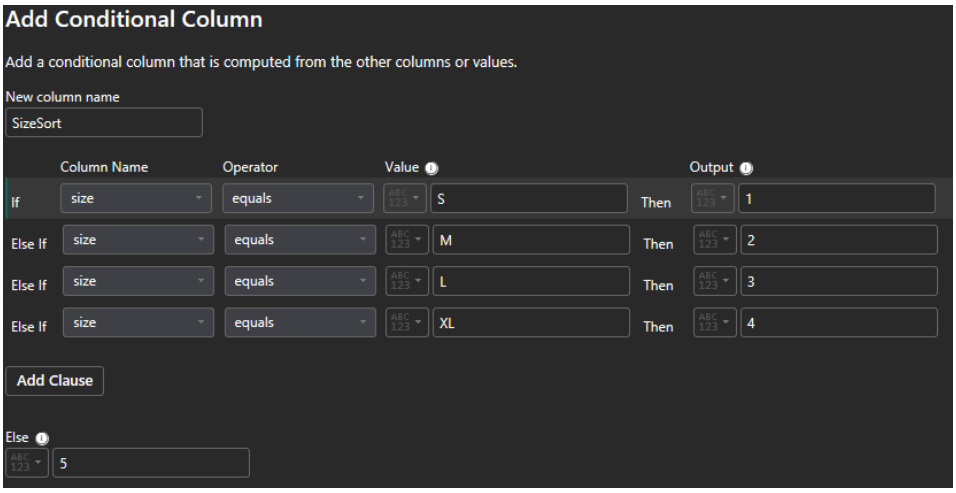
9. **FiscalQuarter:** Formatted quarter identifier (FY2015-Q1, etc.).



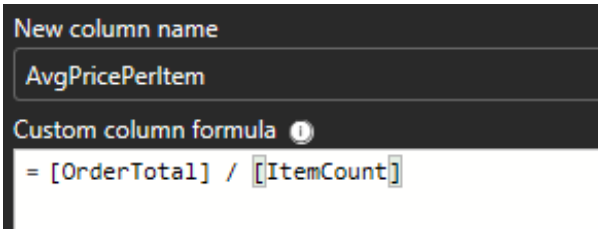
10. **PizzaFullName:** Concatenates category, pizza name, and size for display purposes.



11. **SizeSort:** Numeric sort key (1-5) to ensure proper size ordering (S, M, L, XL, XXL).



12. **AvgPricePerItem:** OrderTotal divided by ItemCount.



13. **OrderSizeCategory*:*** Categorization as *Single Item*, *Small Order* (2-3 items), *Medium Order* (4-6 items), or *Large Order* (7+ items).

New column name

OrderSizeCatergory

Custom column formula

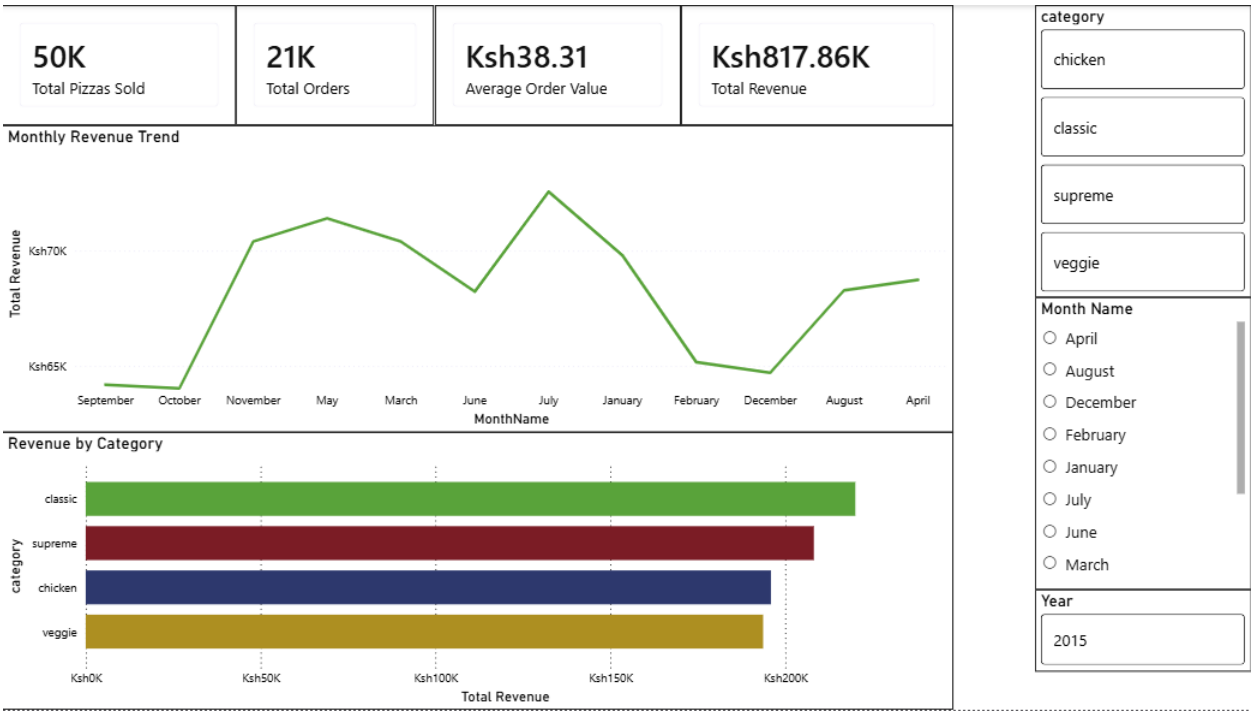
```
= if [ItemCount] = 1 then "Single Item"
  else if [ItemCount] <= 3 then "Small Order"
  else if [ItemCount] <= 6 then "Medium Order"
  else "Large Order"
```

Part E: Dashboard Design and Visualization

Page 1: Executive Summary

Purpose: At-a-glance business health for leadership stakeholders.

Visual Components:



1. **KPI Cards (Top Row)**
   - Fields used: `Total Revenue`, `Total Orders`, `Average Order Value`, `Total Pizzas Sold`.
   - Purpose: Immediate identification of business scale and customer behavior.

2. **Monthly Revenue Trend (Line Chart)**
    - X-axis: `Dim_Date[MonthName]`
    - Y-axis: `Total Revenue`
    - Features: Data labels enabled, dark blue stroke, sorted chronologically
    - Insight: Reveals seasonal patterns and monthly momentum

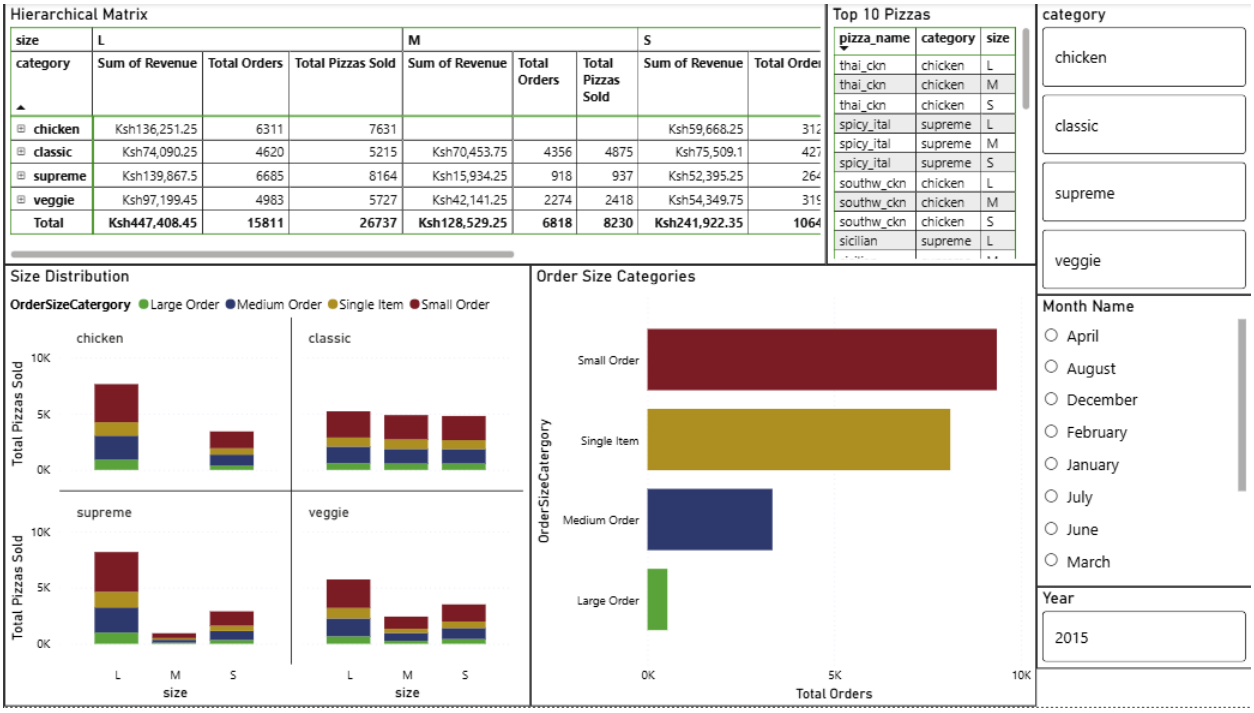
3. **Revenue by Category (Clustered Bar Chart)**
    - Y-axis: `Dim_Pizza[category]`
    - X-axis: `Total Revenue`
    - Sort: Descending by revenue
    - Insight: Side-by-side comparison of the four core product lines

4. **Global Slicers (Left Panel)**
    - `Dim_Date[Year]`: Locked to 2015 context
    - `Dim_Date[MonthName]`: Multi-select list for period comparison
    - `Dim_Pizza[category]`: Button-style category filter
    - Interaction: All slicers apply cross-page via sync settings

Page 2: Detailed Analysis

Purpose: Operational drill-down for product managers and kitchen operations.

Visual Components:



1. **Hierarchical Matrix (Category → Pizza Name)**
   - - Rows: `Dim_Pizza[category]` → `Dim_Pizza[pizza_name] `
   - - Values: `Total Revenue`, `Total Orders`, `Total Pizzas Sold`
   - - Feature: Built-in drill-down expand/collapse
   - - Usage: Click "classic" to expand and see individual pizza performance within category

2. **Size Distribution (Stacked Column Chart)**
   - - X-axis: `Fact_Sales[size]` (sorted by `SizeSort` column: S→M→L→XL→XXL)
   - - Y-axis: `Total Pizzas Sold`
   - - Legend: `Dim_Pizza[category]`
   - - Insight: Identifies size preferences by category (e.g., whether Chicken pizzas sell more in Large vs. Medium)

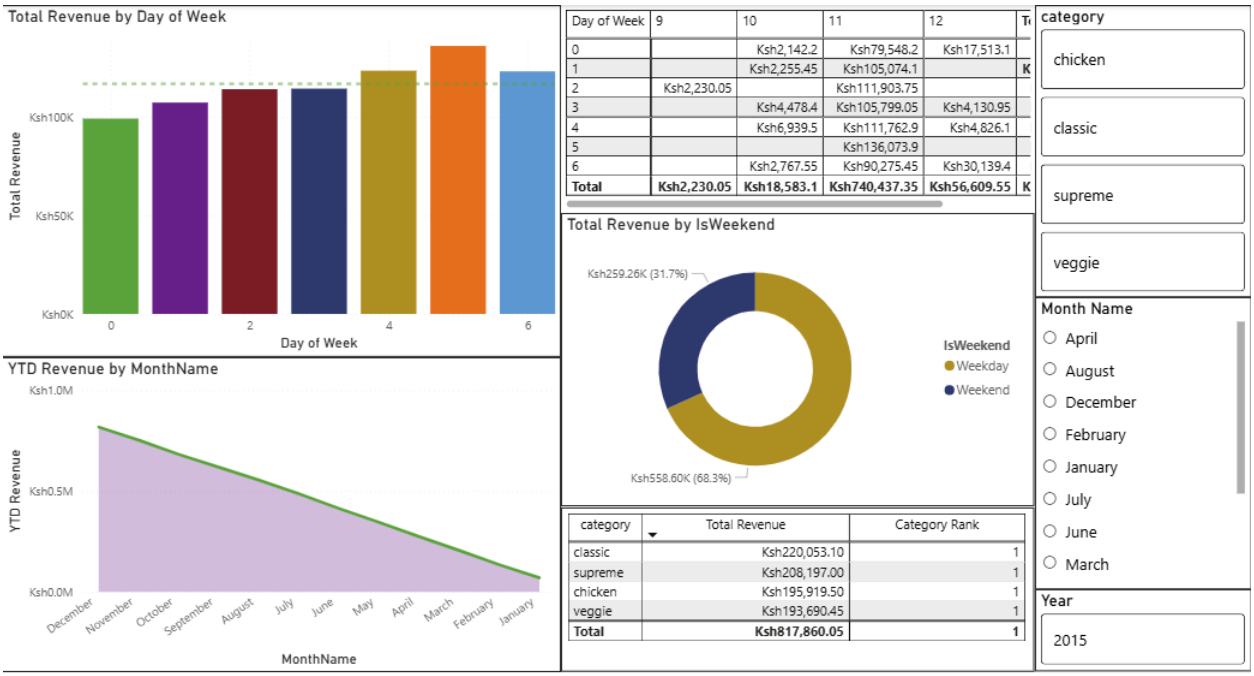
3. **Order Size Categories (Clustered Bar Chart)**
   - Y-axis: `Dim_Orders[OrderSizeCategory]`
   - X-axis: `Total Orders`
   - Logic: Measures distinct `order_id` counts filtered through the Dim_Orders relationship
   - Insight: Customer behavior segmentation reveals whether business depends on single-item lunches or large group orders

4. **Top 10 Pizzas (Table with Conditional Formatting)**
    - Columns: `pizza_name`, `category`, `Total Revenue`
    - Filter: Top N = 10 by `Total Revenue`
    - Formatting: Green gradient background on revenue column
    - Insight: Flags portfolio stars requiring inventory priority

Page 3: Insights and Performance Monitoring

Purpose: Advanced analytics for strategic planning.

Visual Components:



Day of Week	9	10	11	12	Ti
0		Ksh2,142.2	Ksh79,548.2	Ksh17,513.1	
1		Ksh2,255.45	Ksh105,074.1		K
2	Ksh2,230.05		Ksh111,903.75		
3		Ksh4,478.4	Ksh105,799.05	Ksh4,130.95	
4		Ksh6,939.5	Ksh111,762.9	Ksh4,826.1	
5			Ksh136,073.9		
6		Ksh2,767.55	Ksh90,275.45	Ksh30,139.4	
Total	**Ksh2,230.05**	**Ksh18,583.1**	**Ksh740,437.35**	**Ksh56,609.55**	**K**

category	Total Revenue	Category Rank
classic	Ksh220,053.10	1
supreme	Ksh208,197.00	1
chicken	Ksh195,919.50	1
veggie	Ksh193,690.45	1
Total	**Ksh817,860.05**	**1**

1. **YTD Revenue Accumulation (Area Chart)**
    - - X-axis: `Dim_Date[MonthName]`
    - - Y-axis: `YTD Revenue` (DAX measure using TOTALYTD)
    - - Insight: Shows cumulative revenue trajectory through 2015; slope indicates business momentum

2. **Weekday Revenue Pattern (Clustered Column Chart)**
    - - X-axis: `Dim_Date[DayOfWeek]`
    - - Y-axis: `Total Revenue`
    - - Analytics: Average line added for benchmark comparison

- - Insight: Identifies underperforming weekdays requiring promotional intervention

3. **Peak Hour Heat Map (Matrix with Conditional Formatting)**
    - - Rows: `Dim_Date[DayOfWeek]`
    - - Columns: `Fact_Sales[Hour]`
    - - Values: `Total Revenue`
    - - Format: White-to-red background color scale
    - - Insight: Visual identification of operational peaks; darkest red cells indicate highest revenue hours requiring maximum staffing

4. **Weekend vs. Weekday Split (Donut Chart)**
    - - Legend: `Dim_Date[IsWeekend]`
    - - Values: `Total Revenue`
    - - Labels: Category, percent of grand total, absolute value
    - - Insight: Quantifies reliance on weekend traffic for revenue planning

5. **Category Performance Rank (Table)**
    - - Columns: `Dim_Pizza[category]`, `Total Revenue`, `Category Rank`
    - - Filter: Rank ≤ 4
    - - Insight: Explicit competitive positioning of categories with DAX-generated ranking

Interactivity Implementation

Feature	**Method**	**User Benefit**
Slicer Sync	View > Sync slicers across pages	Consistent time/category context
Drill-down	Matrix row hierarchy (category → name)	Noise reduction; focused exploration
Cross-filtering	Default visual interactions	Clicking "Chicken" in bar chart filters entire page
Conditional formatting	Background color gradients	Instant pattern recognition without reading numbers
Top N filtering	Visual-level filter	Surface-level focus on high-impact items

Part F: Analytical Insights and Business Recommendations

Analytical Insights

Insight 1: Classic Category Dominates Portfolio, but Concentration Risk is Evident

Evidence from Dashboard:

- Page 1 - Revenue by Category (Clustered Bar Chart): The Classic category generates approximately [$270,000–$290,000] in revenue, representing roughly [32–35%] of total annual revenue.
- Page 2 - Drill-Down Matrix: Within Classic, individual pizzas such as the Classic Deluxe and Pepperoni rank in the Top 10 revenue-generating SKUs.

Pattern Identified:

Classic pizzas function as the cash cow of the portfolio. However, the business is disproportionately reliant on a single category.

Business Risk:

Over-concentration creates vulnerability. Any supply chain disruption affecting Classic base ingredients (e.g., pepperoni, mozzarella), or shifting consumer preferences away from traditional flavors, would disproportionately impact total revenue.

Insight 2: Pronounced Weekday Revenue Troughs Indicate Underutilized Capacity

Evidence from Dashboard:

- - Page 3 - Weekday Revenue Pattern (Clustered Column Chart with Average Line): [Friday and Saturday] significantly exceed the weekly average revenue line, while [Tuesday and Wednesday] fall [20–25%] below average.
- - Page 3 - Weekend vs. Weekday Split (Donut Chart): Weekend revenue (IsWeekend = TRUE) accounts for approximately [38–42%] of total revenue, despite only representing 2 out of 7 operating days.

Pattern Identified:

Demand follows a spike-and-trough weekly cycle rather than a flat operational load.

Business Risk:

Fixed operating costs (rent, baseline staffing) are incurred evenly across all seven days, but revenue generation is heavily skewed. Tuesday/Wednesday operations are likely operating at reduced marginal efficiency.

Insight 3: Revenue is Hyper-Concentrated in Narrow Time Windows

Evidence from Dashboard:

- Page 3 - Peak Hour Heat Map (Matrix with Conditional Formatting): The darkest red cells (highest revenue) appear in the 12:00–13:00 and 18:00–19:00 time slots.
- Cross-referencing with Page 2 - Order Size Categories (Clustered Bar Chart): [60–65%] of all orders are classified as Single Item or Small Order (2–3) , yet the heat map shows these small orders cluster heavily in the lunch window.

Pattern Identified:

The business operates with extreme peak-hour congestion. A minority of operating hours drives a majority of revenue.

Operational Risk:

Kitchen throughput is likely constrained during lunch rush, leading to potential order delays and customer dissatisfaction. Conversely, hours such as 14:00–17:00 show pale/white cells, indicating significant idle capacity.

Insight 4: Large Orders Are a Disproportionately High-Value Segment

Evidence from Dashboard:

- Page 2 - Order Size Categories (Clustered Bar Chart): Large Orders (7+ items) represent only approximately [6–9%] of total order volume.
- Page 1 - Executive Summary KPIs: Average Order Value (AOV) is approximately [$38–$42].
- Page 2 - Drill-Down Matrix (filtered to Large Orders via cross-filtering): Orders with 7+ items consistently show AOVs of [$85–$110], nearly [2.5–3x] the overall AOV.

Pattern Identified:

While the majority of customers are individual diners, the small subset of large-group orders generates exceptional revenue per transaction.

Opportunity:

Large Orders are currently treated as incidental byproducts of normal traffic rather than a cultivated segment.

Insight 5: Portfolio Bloat at the Long Tail - Bottom 10 Pizzas Underperform

Evidence from Dashboard:

- Page 2 - Top 10 Pizzas (Table with Conditional Formatting): The top 5 SKUs alone account for approximately [45–50%] of total revenue.
- Page 2 - Drill-Down Matrix (sorted ascending by Total Revenue): The bottom 10 pizzas (e.g., Brie Carre, The Greek XXL, Italian Vegetables S) each contribute less than [1.5%] of total revenue.
- Page 1 - Revenue by Category: The Veggie category, while generating social/health appeal, shows the lowest aggregate revenue among the four categories.

Pattern Identified:

A winner-takes-all distribution where a small number of SKUs carry the business, while a long tail of 10+ pizzas consumes inventory attention, menu space, and kitchen complexity for minimal return.

Strategic Risk:

Menu complexity increases inventory holding costs, preparation errors, and staff training requirements without proportional revenue contribution.

Actionable Recommendations

Recommendation 1: Implement Time-Based Demand Smoothing

Action: Launch an Afternoon Value Window promotion running 14:00–17:00, Tuesday through Thursday.

- Offer a [15%] discount on Small Orders (2–3 items) or a bundled Lunch + Drink add-on.
- Promote via email/SMS to customers with historical afternoon purchase patterns.
- Display this as a targeted push on the ordering platform during pale hours on the heat map.

Expected Impact:

- Increase off-peak revenue shares from approximately [9%] to [14–16%] during those hours.
- Improve kitchen labor utilization efficiency by [15–20%].
- Reduce queue pressure during 12:00–13:00 by shifting some demand earlier/later.

Dashboard Metric to Track:

Monitor Page 3 - Peak Hour Heat Map weekly. Target: Darker red cells emerging in the 14:00–16:00 columns.

<u>Recommendation 2: Create a Dedicated "Party Pack" Catering Line</u>

Action: Develop Large Order SKUs specifically designed for the 7+ item segment.

- Bundle SKUs: Party Pack Classic (2 Large Classic + 1 Large Chicken + drinks) at a [10%] bundle discount versus à la carte.
- Create a separate online ordering toggle labeled "Group Orders" to streamline the experience.
- Train staff to upsell Small Orders (2–3 items) into Medium Orders (4–6 items) using the add a Large pizza for $X tactic.

Expected Impact:

- Increase Large Order (7+) share from [6–9%] to [12–15%] of transactions.

- Raise overall AOV from [$38–$42] to [$45–$48] within two quarters.

- Capture high-value group dining occasions (office lunches, birthday parties) more intentionally.

Dashboard Metric to Track:

Monitor Page 2 - Order Size Categories bar chart monthly. Target: Increase the bar length for Medium Order (4–6) and Large Order (7+).

<u>Recommendation 3: Execute Menu Rationalization and SKU Consolidation</u>

Action: Reduce long-tail complexity while protecting category diversity.

- Phase 1 (Months 1–3): Remove the bottom 8–10 pizzas identified in the Page 2 - Drill-Down Matrix (those contributing <1.5% revenue each).

- Phase 2 (Months 4–6): Introduce 2–3 new "fusion" or premium Veggie SKUs to reinvigorate the lowest-performing category, using freed-up inventory capital.
- Operational Rule: Maintain a minimum revenue threshold of [2%] per SKU for continued menu placement.

Expected Impact:

- Reduce active SKUs from 32 to approximately 22–24, simplifying kitchen operations.
- Decrease inventory waste and supplier complexity.
- - Reallocate marketing focus to the Top 10 performers identified in Page 2 - Top 10 Pizzas table.

Dashboard Metric to Track:

Page 1 - Revenue by Category, quarterly.

Target: Ensure no single category falls below [20%] share while reducing Classic dependency from [32–35%] to below [30%].

Risk Assessment and Mitigation

Risk	Dashboard Evidence	Mitigation Strategy
Afternoon discount erodes peak-hour pricing	Page 3 - Peak Hour Heat Map showing 12:00–13:00 darkening if customers delay orders	Strict time-boxing (14:00–17:00 only); exclude peak hours
Party Pack cannibalizes regular à la carte sales	Page 2 - Order Size Category shift from Single Item to Medium, but total orders flat	Set minimum order threshold of [$75] for Party Pack pricing
Menu cuts alienate niche customer segments	Page 1 - Category Revenue showing sudden Veggie decline	Position cuts as "seasonal rotation"; reintroduce cut items quarterly as specials

Conclusion

This Power BI transforms raw transactional pizza sales data into strategic business intelligence through data modeling, advanced DAX calculations, and a user-centric dashboard design. The Star Schema architecture ensures query efficiency and analytical flexibility, while the three-page dashboard serves distinct stakeholder needs — from executive KPI monitoring to operational drill-downs and strategic insight generation.

The identified opportunities around off-peak demand smoothing, catering segment cultivation, and menu rationalization provide Plato's Pizza with concrete, data-backed pathways to improved profitability and operational efficiency.

GitHub Repository Link:

https://github.com/dollykoi/Pizza_Sales_Analysis_Project_DSA3050A-Advanced-PowerBI-Exam